

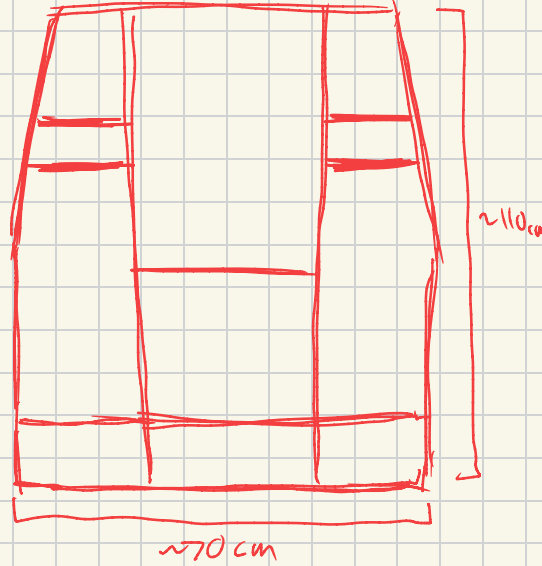
Dough - Kust



Kaden
Ship
7/22/26



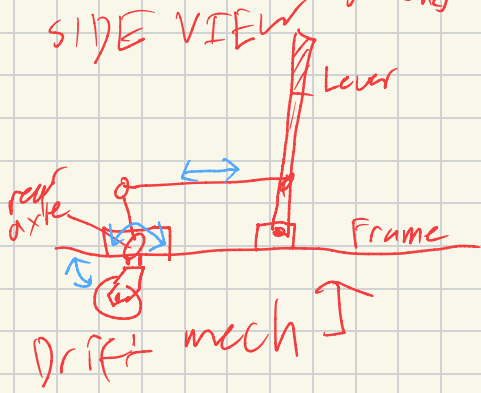
Main frame
(Aluminum? steel?)



Lever front = all casters pos. down?
Lever pulled back - back casters + powered wheel on ground



SIDE VIEW



NOTE: this is just the Razor (m2) cart

wheels not touching



Potential add-ons:

- NFC chip in kart → use phone as key?
- Control whole thing with Wiimote?
- Spark generator? flame?



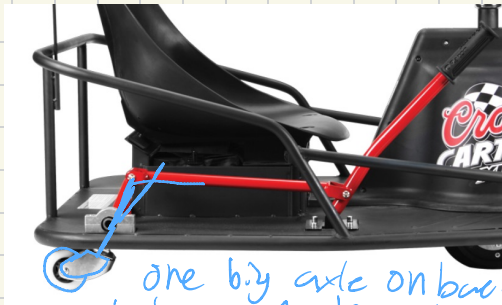
Mario kart controls

Timeline:

1. CAD rough model of kart to ensure drift mech. works (print + test)

2. CAD kart subsection by subsection

- Chassis
- powered wheel module
- Casters
- Drift mechanism
- Electrical
- seat + controls
- Braking
- Guards/cosmetic add-ons

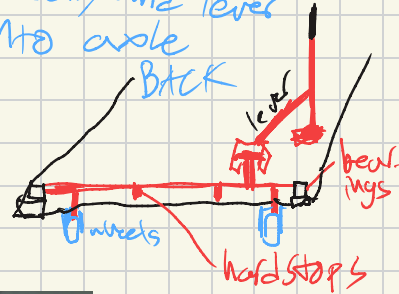


one big axle on back
hard stop, wheels, and lever
welded onto axle



hard stop

bearing block



Hub motor v.s. chain drive

PROS

CONS

hub	chain	hub	chain
• quieter • less maintenance • less parts • better power transfer • more compact • simpler PWM (powered wheel mob)	• cheaper (already have wheel) • better torque • adjustable gear ratio • allows for better suspension • more efficient?	• longer of two low/high torque • significantly more expensive • Virtually no suspension, I feel every bump	• more parts = high chance things break/hot to repair • chain maintenance needed • more space needed in the PWM

Back axle material: 1.000 schedule 40 + handle

Frame material: mild steel

Back axle mod: single socket tees for casters

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• Figuring out F360, not bad so far

• Decided to do a single lever for back axle for simplicity

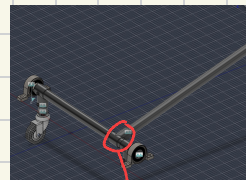
• To reduce # of welds, decided to use socket tees for lever wheels

◦ Using through-bolts instead of the setscrew for strength

• Deciding bwn expanding stem casters or threaded stem casters w/ threaded insert

◦ expanding is more convenient, but weaker

◦ threaded would req. more parts + welding



single socket tee →



8/11/26 (cont.)

• Also decided to use an e-scooter fork for the PWM instead of trying to weld one myself

Fig1.

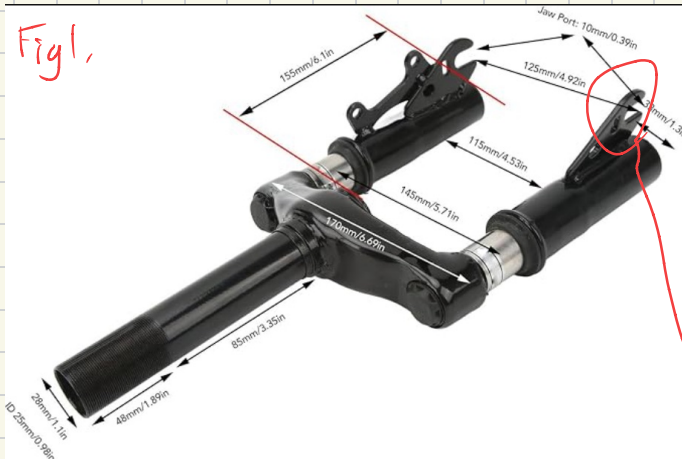


Fig2



- o suspension
- o smoother steering
- o requires some drilling of the dropouts (see fig1)
- o expensive (\$)
- o Can use head tube / part of frame from my old Razor e200 (Fig.2)
 - Cut off part and weld to frame???

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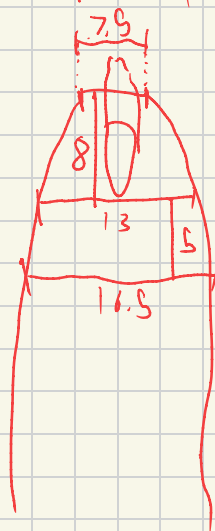
• Drift mech: 1.000" sched. pipe, single socket tees w/ thru-bolts, expanding steam custers w/ well nuts for vibration reduction

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Razer E200 dimensions



TOP (cm)



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decision abt frame: 1.25x1.25x

0.083 structural sq. tube

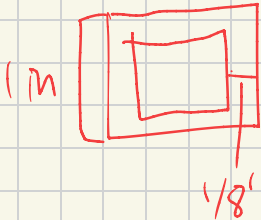
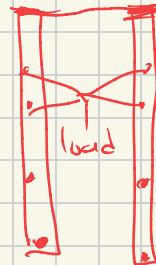
drift mech: 1.00x0.045 4026 round
tube

8/24 UPDATE

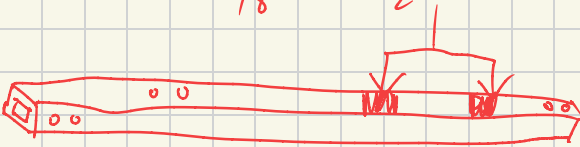
using 6061/6063 ALUMINUM instead, but need
to do FEA to see if it will hold.

SIDE

Load (secant)



Two beams side by side:
holds 2x load
 $\frac{1}{2}$ load (2 beams)



Will 6063 Alu hold?

Force: $m \cdot g$

$m: 150 \text{ lb} \approx 68 \text{ kg}$

$g = 9.8 \text{ m/s}^2$

Correction: forget mass of chair, butt, etc

assume $m \approx 200 \text{ lb}$

$$\frac{1}{2}F = (200 \cdot 9.8) \times 0.5 = 980 \text{ N}$$

$$\frac{1}{2}F = (68 \times 9.8) \times 0.5 \approx 333 \text{ N}$$

Factor of Safety: 2.89

6063-T52 Alu

